Companion XC: End-to-End RDCC Inference Pipeline — From Simulations to Posteriors

Michael Lehmann

mi.lehmann@gmx.de

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Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces the relaxation parameter α, which modifies the scale-dependent gravitational potential and induces observable signatures in weak-lensing convergence fields. Previous Companions developed topological transport statistics, multi-probe likelihoods, neural surrogates, hierarchical Bayesian inference, and emulator-accelerated pipelines. In this work, we assemble these components into a unified end-to-end RDCC inference pipeline. The pipeline begins with simulations, extracts multi-probe summary statistics, evaluates neural joint likelihoods, incorporates emulator uncertainty, applies hierarchical Bayesian modeling, and produces final posteriors for α. This Companion provides the complete computational and statistical workflow required for applying RDCC to Euclid-like survey data.

1 Pipeline Overview

The end-to-end RDCC inference pipeline consists of the following stages:

1. Simulation of weak-lensing convergence maps for a grid of α.
2. Extraction of multi-probe summary statistics:
   - transport-based persistence-diagram statistics,
   - Minkowski functionals,
   - shear power spectra.
3. Training of neural joint likelihoods (Companion LXXXVI).
4. Calibration via flow-matching and score diagnostics (Companion LXXXVII).
5. Emulator acceleration using neural transport surrogates (Companion LXXXVIII).
6. Hierarchical Bayesian inference with transport-regularized posteriors (Companion LXXXIX).
7. Posterior sampling and uncertainty quantification.

2 Simulation Stage

We generate weak-lensing convergence maps $\kappa(\theta)$ for a grid of α values using:

- N-body simulations,

- fast approximate solvers,
- multi-fidelity simulation flows (Companion LXXXVIII).

Each simulation produces:

- convergence maps,
- persistence diagrams,
- Minkowski functional curves,
- shear power spectra.

3 Summary-Statistic Extraction

We compute:

- transport distances $\mathcal{T}_\ell$,
- Minkowski functionals $V_i(\nu)$,
- shear power spectra $C_\ell^{\gamma\gamma}$.

These form the multi-probe summary vector:

$$\mathbf{s} = \left\{\mathcal{T}_\ell, V_0(\nu), V_1(\nu), V_2(\nu), C_\ell^{\gamma\gamma}\right\}.$$

4 Neural Joint Likelihood Evaluation

We evaluate the neural joint likelihood $\mathcal{L}_\theta$ from Companion LXXXVI:

$$\mathcal{L}_\theta(\mathbf{s} \mid \alpha) = \mathcal{N}(f_\theta(\mathbf{s}); \mu(\alpha), \Sigma(\alpha)) \left|\det \frac{\partial f_\theta}{\partial \mathbf{s}}\right|.$$

Calibration ensures statistical correctness (Companion LXXXVII).

5 Hierarchical Bayesian Inference

We define the hierarchical posterior:

$$p(\alpha, \theta_{\text{emu}} \mid \mathbf{s}_{\text{obs}}) \propto \mathcal{L}_\theta(\mathbf{s}_{\text{obs}} \mid \alpha, \theta_{\text{emu}}) p(\theta_{\text{emu}}) p(\alpha).$$

Transport regularization (Companion LXXXIX) enforces geometric consistency.

6 Posterior Sampling and Final Constraints

We sample the posterior using:

- Hamiltonian Monte Carlo,
- NUTS,
- variational inference,
- neural posterior estimation.

The final output is:

$$p(\alpha \mid \mathbf{s}_{\rm obs})$$

with credible intervals and diagnostic checks.

7 Conclusion

This Companion assembled all previous components into a unified end-to-end RDCC inference pipeline. The pipeline integrates simulations, multi-probe summary statistics, neural likelihoods, emulator acceleration, hierarchical Bayesian modeling, and transport regularization. It provides a complete, scalable, and statistically robust workflow for inferring the RDCC relaxation parameter α from Euclid-like weak-lensing data.

References

[Kilbinger(2015)] M. Kilbinger, Reports on Progress in Physics **78**, 086901 (2015).

[Bartelmann and Schneider(2001)] M. Bartelmann and P. Schneider, Physics Reports **340**, 291 (2001).

[Heitmann *et al.*(2009)Heitmann *et al.*] K. Heitmann *et al.*, Astrophysical Journal **705**, 156 (2009).

[Lawrence *et al.*(2017)Lawrence *et al.*] E. Lawrence *et al.*, Astrophysical Journal **847**, 50 (2017).

[Knabenhans *et al.*(2021)Knabenhans *et al.*] M. Knabenhans *et al.*, Monthly Notices of the Royal Astronomical Society **505**, 2840 (2021).

[Cranmer *et al.*(2020)Cranmer, Brehmer, and Louppe] K. Cranmer, J. Brehmer, and G. Louppe, Proceedings of the National Academy of Sciences **117**, 30055 (2020).

[Greenberg *et al.*(2019)Greenberg, Nonnenmacher, and Macke] D. Greenberg, M. Nonnenmacher, and J. Macke, International Conference on Machine Learning , 2404 (2019).

[Jeffrey *et al.*(2021)Jeffrey, Alsing, and Lanusse] N. Jeffrey, J. Alsing, and F. Lanusse, Monthly Notices of the Royal Astronomical Society **501**, 954 (2021).

[Hahn *et al.*(2022)Hahn *et al.*] C. Hahn *et al.*, Nature Astronomy **6**, 1185 (2022).

[Papamakarios *et al.*(2019)Papamakarios, Nalisnick, Rezende, Mohamed, and Lakshminarayanan] G. Papamakarios, E. Nalisnick, D. J. Rezende, S. Mohamed, and B. Lakshminarayanan, arXiv:1912.02762 (2019).

[Rezende and Mohamed(2015)] D. J. Rezende and S. Mohamed, International Conference on Machine Learning , 1530 (2015).

[Brehmer and Cranmer(2020)] J. Brehmer and K. Cranmer, Advances in Neural Information Processing Systems **33**, 442 (2020).

[Ramesh *et al.*(2022)Ramesh *et al.*] P. Ramesh *et al.*, arXiv:2210.02747 (2022).

[Gelman *et al.*(2013)Gelman *et al.*] A. Gelman *et al.*, CRC Press (2013).

[Betancourt(2017)] M. Betancourt, arXiv:1701.02434 (2017).